FIG. 3 — SIX-DOMAIN MERKLE FOREST 300

CREDENTIAL COMMITMENT 302

```
credentialCommitment = Poseidon.hash([credentialType, forestRoot, issuerWalletId,
                              subjectWalletId, issuedAt, expiresAt, revoked])
(7 Fields — single hash embedded in credential)
```

FOREST ROOT COMPUTATION 304

```
forestRoot = Poseidon.hash([root_0, root_1, root_2, root_3, root_4, root_5])
(6 domain tree roots combined into single Field)
```

SIX DOMAIN TREES (MerkleTree height=8, 128 leaves each) 306

TREE 0: MEMBER 308	TREE 1: PRESCRIBER 310	TREE 2: PHARMACY 312
Domain Tag: 0 Leaf: H(cardholderIdHash, dateOfBirth, genderCode, firstNameHash, lastNameHash)	Domain Tag: 1 Leaf: H(prescriberIdQual, prescriberIdHash)	Domain Tag: 2 Leaf: H(serviceProviderIdQual, serviceProviderIdHash)
TREE 3: MEDICATION 314	**TREE 4: ACCESS_CTRL 316**	**TREE 5: TRANSACTION 318**
Domain Tag: 3 Leaf: H(rxRefNumberHash, ndcHash, quantityDisp, fillNumber, daysSupply, compoundCode, dawCode)	Domain Tag: 4 Leaf 0: EligibilityData Leaf 1 : PriorAuthData Leaf 127: AccumulatorData (deductible, oop, max)	Domain Tag: 5 Leaf: H(fulfillment .commitment(), patientPay, planPay)

TREE DETAIL — ACCESS CONTROL 316 (Expanded) 320

Leaf Index	Contents	Structure
0	EligibilityData 320-A	H(planIdHash, groupIdHash, effectiveDate, terminationDate, copayAmountCents, coverageType, formularyRoot, networkRoot)
1 .. 126	PriorAuthData 320-B (per drug)	H(ndcHash, memberWalletId, prescriberWalletId, approvalDate, expirationDate, approvedQuantity, authCodeHash)
127	AccumulatorData 320-C	H(planYear, deductibleAmountCents, deductiblePaidCents, oopMaxCents, oopPaidCents, annualBenefitMaxCents, annualBenefitUsedCents)

SELECTIVE DISCLOSURE PROTOCOL 330

```
Step 1 (332): Prover selects leaf index i in domain tree t
Step 2 (334): Prover computes MerkleWitness(tree_t, i) — sibling path, height 8
Step 3 (336): Prover supplies leaf preimage fields as private inputs to ZkProgram
Step 4 (338): Verifier checks: H(preimage) == leaf_i, witness.calculateRoot() == root_t,
                H(...all roots...) == forestRoot, forestRoot matches credential

Result: Verifier learns ONLY the disclosed fields. All other trees/leaves stay hidden.
```

COMPOSITION CHAIN (Credential to Settlement) 340

Level	Ref	Proof	Proves
L0	342	EligibilityProof	Member is eligible under benefit plan
L1	344	PrescriptionProof	Prescriber authorized for this medication
L2	346	FulfillmentProof	Pharmacy dispensed, copay/planpay computed
L3	348	AccumFulfillmentProof	Accumulators updated (deductible, OOP)
L4	350	StateTransitionProof	Forest root updated, inline wallet verify
L5	352	BatchProof/ChainProof	Sequencer batches into chain (FIG. 2)

MERKLE FOREST STRUCTURE (Visual) 360

